

Kateryna Morhun

Cambridge, MA | [LinkedIn](#) | [Github](#) | katerynamorhun.com | kmorhun@mit.edu

Education

Massachusetts Institute of Technology (MIT)

Cambridge, MA

- BS, MEng in Computer Science; AI and Society Concentration BS 2025; MEng 2026 (GPA 5.0)
- Coursework: Advanced NLP, Applied Statistics, AI Decision Making & Society, Data Visualization

Work Experience

Cadence Care

June 2026-August 2026

Software Engineer

Remote

- Deployed an AI voice agent with ElevenLabs to manage appointment rescheduling for 180k+ patients
- Developed robust evaluations of LLM agent safety, reliability, and edge case management

MIT Course: Data Visualization & Society

January 2026-June 2026

Teaching Assistant

Cambridge, MA

- Designed and delivered course curricula introducing students to HTML, CSS, JS, Svelte, and D3.js
- Mentored students (n=112) in weekly office hours and student-facing discussion forums
- Co-designed assignment rubrics and graded assignments weekly

Cambridge Mobile Telematics

June 2025-August 2026

Software Engineer I, Cloud

Cambridge, MA

- Scaled serverless ETL pipeline processing 200+ GB of telematics data weekly (Redshift Spectrum→DynamoDB)
- Designed and deployed full AWS cloud infrastructure via Terraform
- Engineered space-efficient solutions for temporal speed distribution analysis
- Built Tornado ECS web app for quantile-based speeding detection
- Delivered technical talks and insights to leadership

Johns Hopkins Center for Language and Speech Processing (CLSP)

June 2024-May 2025

Software Engineer, ML Engineer, AI-curated Democratic Discourse

Baltimore, MD

- Consistently met project deadlines by translating research ideas into actionable development steps
- Rapidly prototyped a human annotation dataset creation pipeline in 48 hours
- Aligned LLM annotations of social media posts to human annotations to improve model reliability
- Developed a recursive, heap-based algorithm for balancing diversity and quality of social media post stacks
- Designed and executed an intuitive 15+ minute technical talk to 60+ non-expert stakeholders
- Fostered interdisciplinary collaboration and grounded technical project components in sociological origins
- Facilitated effective communication, dependency management, and knowledge transfer across subteams

MIT CS+AI Lab - Decentralized Information Group

February 2024-May 2024

Researcher

Cambridge, MA

- Researched & cataloged 100+ AI legislation & auditing requirement documents

City Form Lab

September 2023-January 2024

Researcher - Tile2Net: pedestrian infrastructure mapping

Cambridge, MA

- Incorporated feedback from stakeholders and global urban planners into the engineering process
- Committed 4000+ lines of detailed documentation on an open-source computer vision tool
- Developed 3 ways to run the tool, including Google Colab and High Performance Clusters
- Prepared 200+ sq mi of GIS tile input data to the model

US Department of Transportation - Volpe Center

June 2023-August 2023

Data Scientist

Cambridge, MA

- Modernized and developed full-stack GIS webapps in ESRI ecosystems to ease equitable transportation policy
- Rapidly prototyped a 10-20x speedup for generating “high injury networks” with a novel network analysis approach
- Analyzed and condensed 5GB+ US Census Data via Pandas and Arcpy to extract actionable insights
- Elicited \$10k+ from sponsors for future development with persuasive technical communication

MIT Course: Differential Equations

February 2023-May 2023

Teaching Assistant

Cambridge, MA

- Mentored students (n=10) in weekly office hours to support their studies

FindOurView

May 2022-August 2022

Full-Stack Developer

Remote

- Refactored frontend from Vue 2.0 to Vue 3.0
- Facilitated dozens of Django and PostgreSQL database migrations in 3 novel site features
- Implemented over 10% increase in Selenium CI/CD test script efficiency and coverage

Digital Humanities Lab

September 2021-May 2022

Researcher - Gender Analysis Toolkit; Self-Sustaining Cities

Cambridge, MA

- Assured RESTful Django API for word distinctiveness analysis algorithms
- Compiled 20+ city documents and primary sources about community-building in self-sustainable black neighborhoods
- Presented to 30+ fundraisers about my research

Leadership/Other Experience

Responsible Tech University Network Board of Directors

September 2024-September 2025

MIT EECS | Morais and Rosenblum Undergraduate Research and Innovation Scholar

September 2024-May 2025

Ukraine Ministry of Education Scholar Support Office

February 2023-August 2023

- Facilitated development of an online hub for Ukrainian displaced scientists, to be presented to the Ministry of Education

Deep Learning; Data-Centered AI (Intro Classes)

January 2023

1st place competition winner

Cambridge, MA

- Reduced bias and uncertainty in risk-aware models using the Capsa and Cleanlab libraries
- Built, trained, and improved deep learning models to generate music and recognize faces using TensorFlow

Ukraine@MIT

February 2022-Present

Core Officer

Cambridge, MA

- Facilitated sustainable community-building for 40+ students
- Trained and mentored underclassmen in leadership skills and institutional knowledge

MIT Ballroom Dance Team

May 2022-Present

President/Captain, Treasurer, Webmaster, Funding Coordinator

Cambridge, MA

- Coordinated 60+ dancers and 8 coaches in reviving a nationally known ballroom competition with 700+ competitors
- Managed \$200,000+ in team assets, expenses, revenue, and donations to maintain sustainable operations
- Reignited intrateam community to pre-COVID-19 levels

Awards/Grants

J. Francis Reintjes Excellence in 6A Industrial Practice Award 2026

4x grant winner (\$15,500) - MIT Council of the Arts (CAMIT) 2022, 2023, 2024, 2025

5x grant winner (\$9,800) - MIT Large Events Fund (LEF) 2021, 2022, 2023, 2024, 2025

1st place winner - MIT Deep Learning Competition with ThemisAI 2023

Top 10% entries - Khanacademy Breakthrough Junior Challenge 2020

Skills/Interests

Technical: Agentic AI, Voice Agents, Model Context Protocol, Terraform, Go, Serverless Cloud Architecture, AWS, Redis, DynamoDB, Redshift Spectrum, RESTful APIs, Tornado, Docker, Python, PyTorch, TensorFlow, Numpy, Pandas, ESRI ArcGIS, Typescript, Javascript, CSS, HTML, C, Vite, Django, Git, SQL, Vue, React

Languages: English (Fluent), Ukrainian (Fluent)

Personal: Quick learner, Easily connect with people, Technical communication, Project management

Interests: Pre-champ competitive ballroom dancing, Anatomically accurate crochet, Dungeons & Dragons